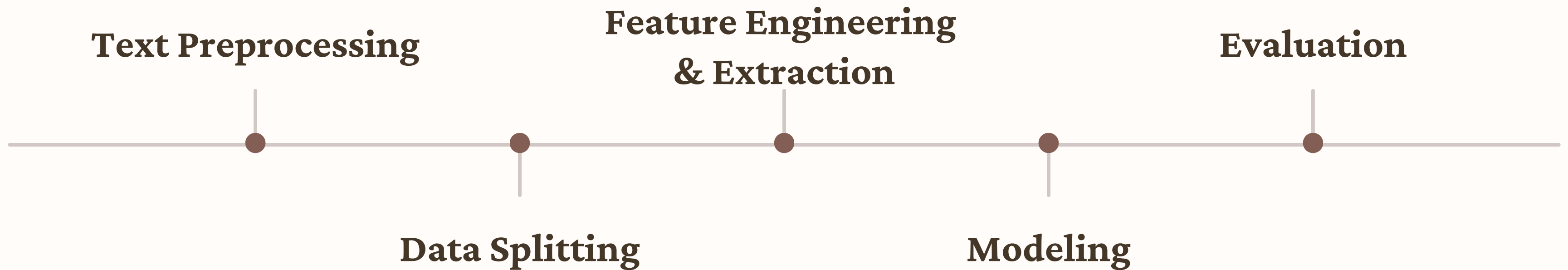




ML FOR TEXT MINING

Section 4

The Standard Pipeline



The Standard Pipeline

Modeling

An individual learner (or **Base Learner**) is a standalone algorithm, such as a SVM, trained to perform a specific task.

Individual learners are fast to train and easy to interpret, but they are prone to specific errors based on their mathematical architecture.

Individual models fail to generalize:

- **High Bias (Underfitting):** Simple models might miss the underlying complexity of the data (e.g., a straight line trying to fit a curved relationship).
- **High Variance (Overfitting):** Complex models might "memorize" the noise in the training set, making them fail when they see unseen data.
- **The Instability Factor:** Small changes in the training data can lead to drastically different predictions in individual models like Decision Trees.

Introducing the Ensemble Learning

Objective: Shifting from a single predictor to a "Wisdom of the Crowd" approach.

Ensemble Learning is the strategic combination of multiple base learners to produce one optimal predictive model.

By aggregating the outputs of many "Weak Learners", we create a "Strong Learner" that is more robust and accurate than any of its individual parts.

Ensemble Learning Methods

Voting

Running different algorithms in parallel and choosing the most frequent result (The Democracy approach).

Bagging

Running the same algorithm on different "random slices" of data to reduce variance (The Diversity approach).

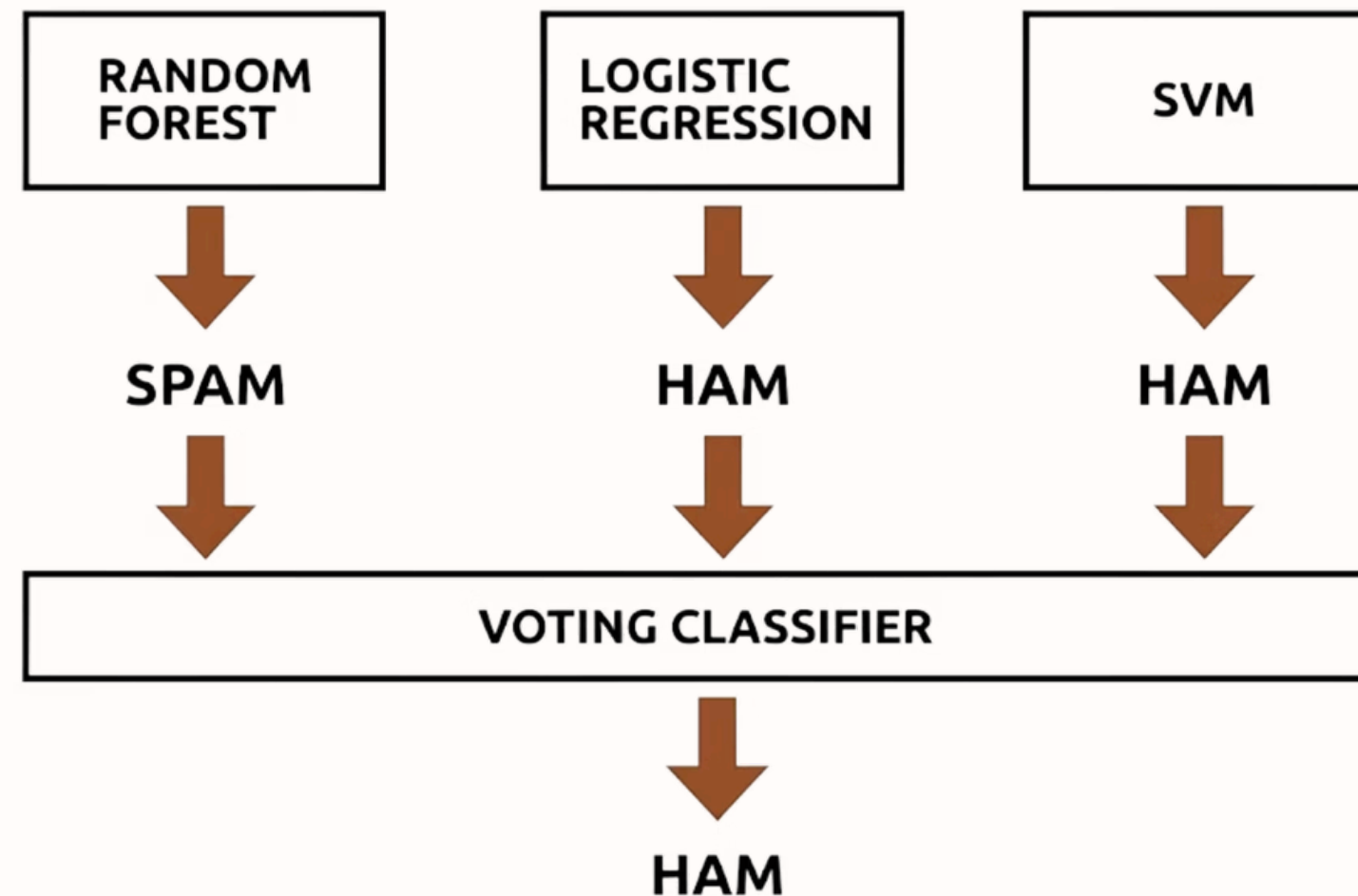
Boosting

Training models one after another, where each new model focuses on the mistakes of the previous one (The Improvement approach).

Ensemble Learning Methods

Voting Method

In a Voting ensemble, we train several **different types of models** (e.g., a Logistic Regression, a SVM, and a Random Forest) on the same dataset. Instead of relying on just one, we let them "vote" on the outcome.

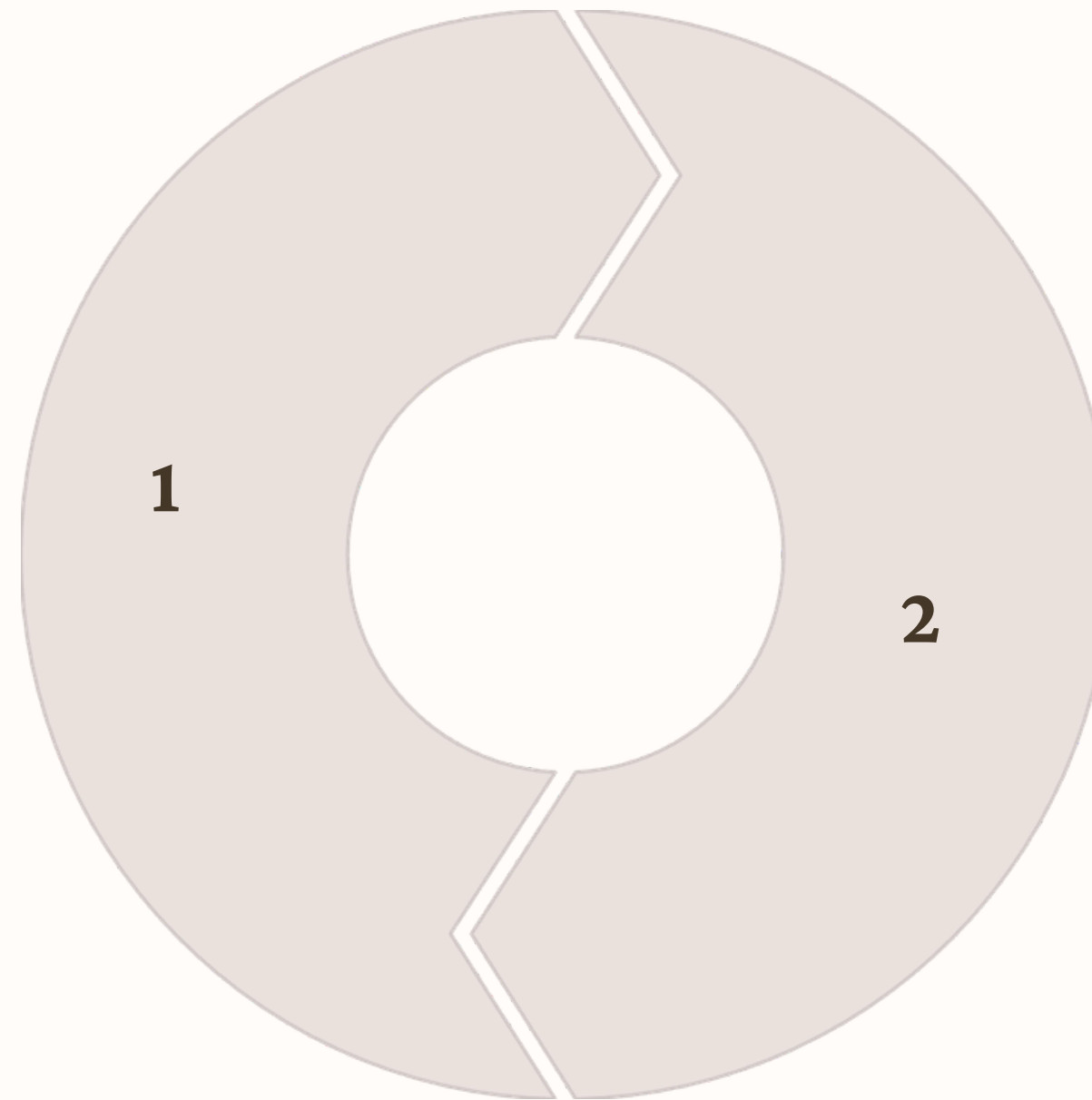


Ensemble Learning Methods

Voting Method

Hard Voting (Majority Rule)

The class that receives the **majority of votes** becomes the ensemble's prediction.



Soft Voting (Weighted Probabilities)

Averaging the **predicted probabilities** for each class across all models. The class with the highest average probability wins.

Ensemble Learning Methods

Voting Method: Scenario

Suppose we have 3 models predicting the class of a single email (Spam vs. Ham).

Model	Probability for Ham	Probability for Spam
Model 1	0.49	0.51
Model 2	0.49	0.51
Model 3	0.90	0.10

Hard Voting (Majority Rule)

Model 1 predicts: Spam ($0.51 > 0.49$)

Model 2 predicts: Spam ($0.51 > 0.49$)

Model 3 predicts: Ham ($0.90 > 0.10$)

Final Ensemble Prediction: Spam (2 votes for Spam, 1 vote for Ham)

Soft Voting (Average Probability)

Average Probability for Ham: $(0.49 + 0.49 + 0.90) / 3 = 0.626$

Average Probability for Spam: $(0.51 + 0.51 + 0.10) / 3 = 0.373$

Final Ensemble Prediction: Ham

Ensemble Learning Methods

Voting Method: Limitation

In the **Voting** method, we relied on **Algorithm Diversity**. We used different "perspectives" to look at the same data.

Finding different algorithms that all perform well on specific dataset can be difficult and time-consuming to tune.

What if, instead of using different types of models, we use the **same** powerful algorithm, but trained it on different "versions" of our dataset?

This is the core logic of **Bagging**.

Ensemble Learning Methods

Bagging Method

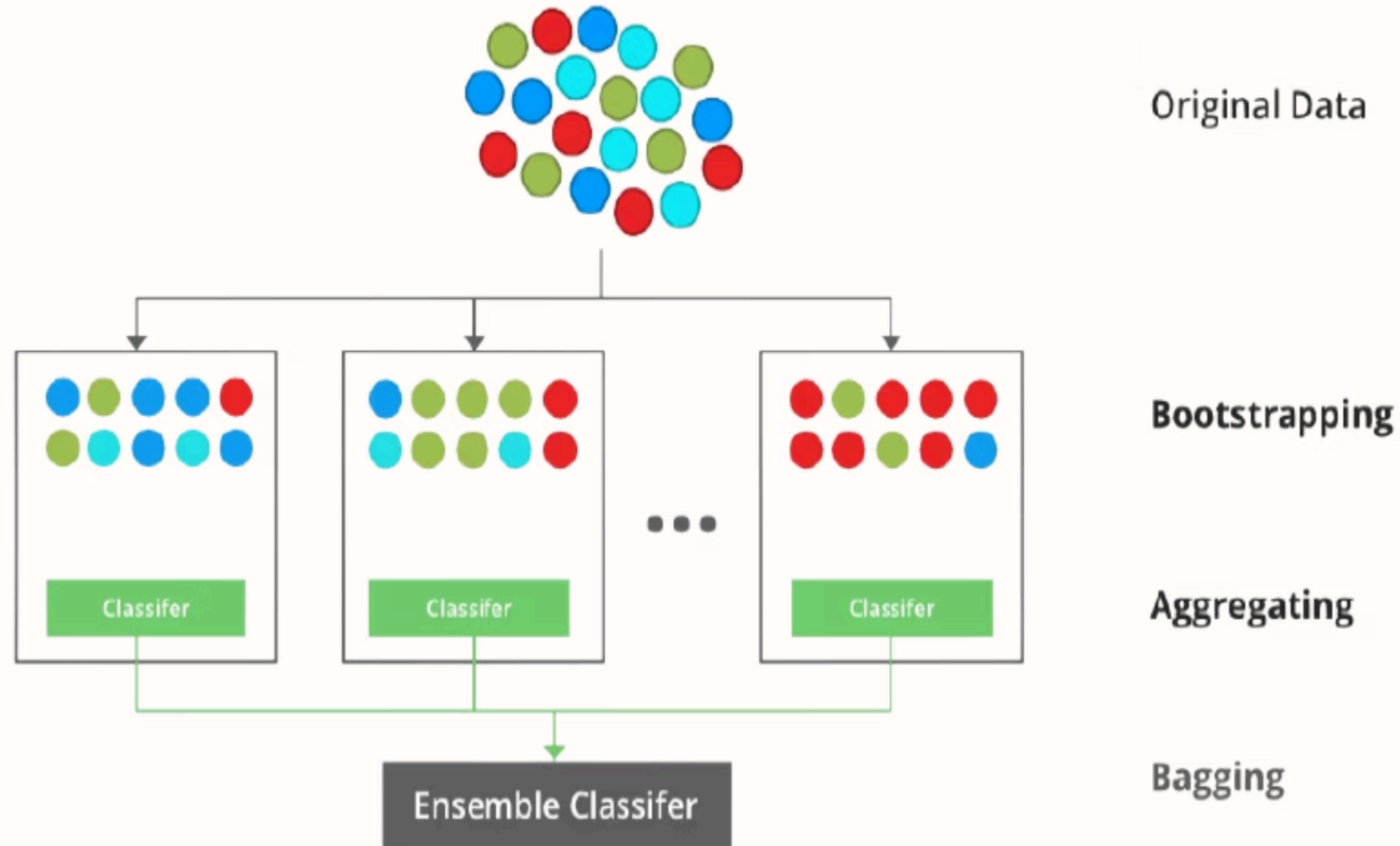
Bagging (Bootstrap Aggregating) is a method designed to reduce **Variance** (the tendency of a model to overfit).

We create multiple subsets of our original training data using a technique called **Bootstrapping**: sampling **with replacement**.

When a training instance is selected for a subset, it is "placed back" into the original dataset, making it available to be selected again for the **same subset**. Some data points may appear multiple times, while others may be excluded.

Ensemble Learning Methods

Bagging Method



Ensemble Learning Methods

Bagging Method

The mechanics that make Bagging a robust ensemble strategy:

- **Diversity through Randomness:** By training each model on a slightly different version of the data, we ensure the models are **de-correlated**. If one model makes a mistake on a specific outlier, the others likely will not, canceling out the error.
- **The Aggregation Step:** Once the individual models are trained, we combine their results:
 - **Classification:** Uses **Majority Voting** (the most frequent class wins).
 - **Regression:** Uses the **Average** of all predicted values.
- **Out-of-Bag (OOB) Evaluation:**
 - The training instances ,that are never sampled for any given predictor, are called **OOB instances**.
 - **The Benefit:** We can evaluate the ensemble's performance using these points during training, effectively getting a validation score without setting aside a separate test set.

Ensemble Learning Methods

Bagging Method: Pasting - The Alternative

Pasting is functionally identical to Bagging, but it uses **Sampling Without Replacement**.

Once a data point is selected from the "bag" for a specific model, it is removed. It cannot be picked again for that same model's training subset (no duplicates in a single model's subset).

Why use Pasting?

- **Best for Large Datasets:** When you have millions of data points, you can give each model a unique, non-overlapping slice of data.
- **Computational Efficiency:** Sometimes faster than Bagging since it doesn't have to manage the "replacement" logic or handle the overhead of repeated data.

Ensemble Learning Methods

Bagging vs. Pasting

Feature	Bagging	Pasting
Primary Goal	Reduce Variance (Overfitting)	Reduce Variance (Overfitting)
Training Style	Parallel (Simultaneous)	Parallel (Simultaneous)
Sampling Rule	With replacement	Without replacement
Data Diversity	High (due to duplicates/missing)	Lower (fixed unique slices)
Dataset Size	Best for Small/Medium	Best for Large

Ensemble Learning Methods

The Shift in Strategy

Bagging & Pasting are great at smoothing out over-complex models with **High Variance**.

But what if your base model is **too simple**?

The Solution is Boosting: While Bagging targets **Variance**, Boosting is specifically designed to deal with **Bias (Underfitting)**.

We shift from **Parallel** (independence) to **Sequential** (collaboration).

Ensemble Learning Methods

Boosting Method

Boosting refers to any Ensemble method that can combine several **weak learners** into a **strong learner**.

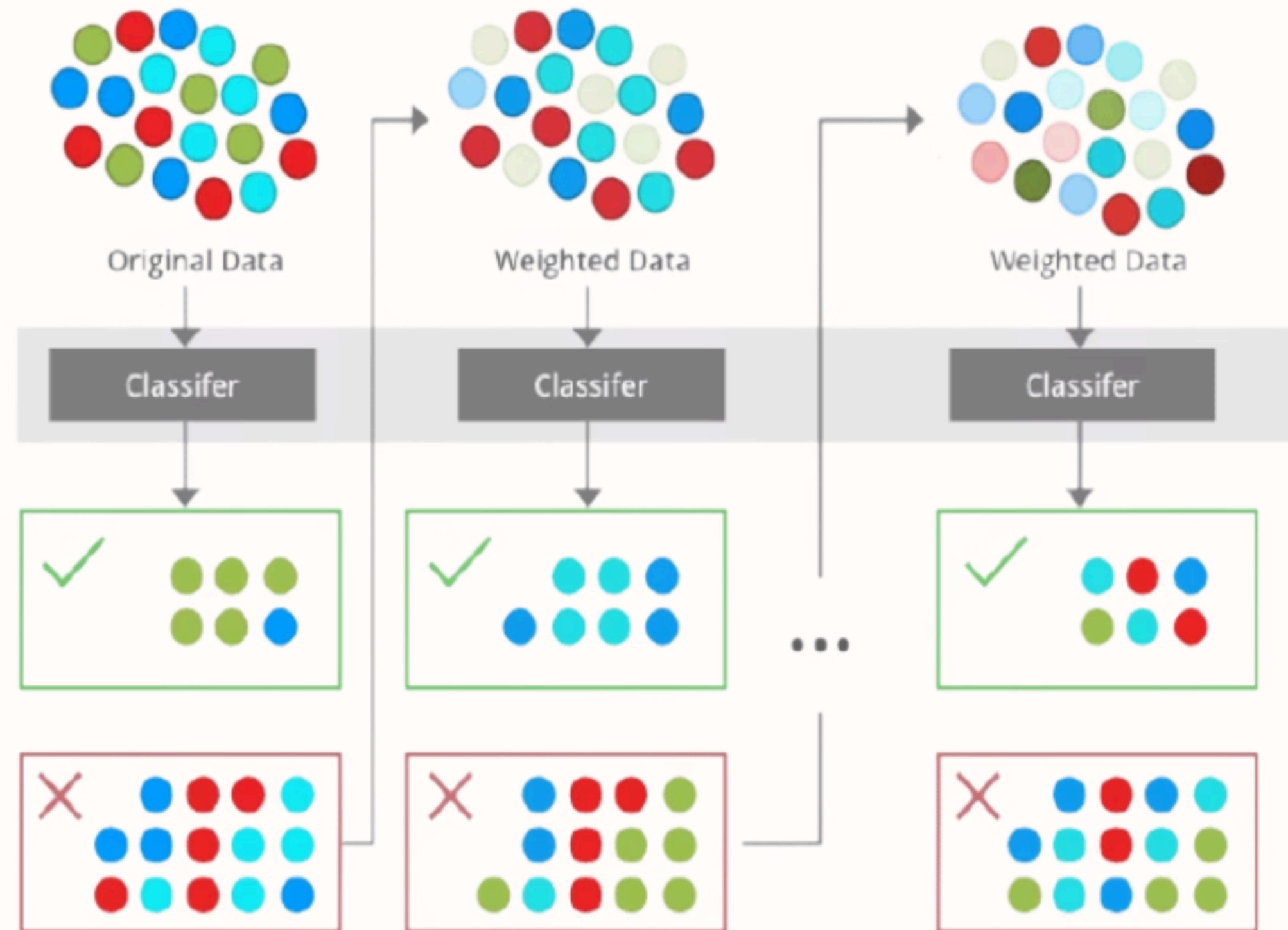
Unlike Bagging, which trains models in parallel, Boosting trains models **sequentially**, each trying to correct its predecessor.

It focuses on the "hard" cases. Data points that were misclassified by Model 1 are given higher priority (or higher weight) when training Model 2.

To systematically reduce **Bias**. It takes a model that is "too simple" and makes it progressively more accurate.

Ensemble Learning Methods

Boosting Method



Ensemble Learning Methods

Boosting Method: Weight-based Approach

We start with a **Weak Learner** (this is often a **Decision Stump: Decision Tree with depth = 1**).

1. **Assign Equal Weights:** Start by giving every data point in your dataset an equal weight (e.g., $1/N$).

2. **Train The Weak Learner**

3. **Calculate Model Importance:** Calculate how accurate this model was.

- A more accurate model gets a higher Amount of Say (α) in the final vote.
- If the model is very accurate (low error), it gets a **high α** (a loud voice).
- Total Error is weighted sum of the current weights of only those incorrect instances.

$$\epsilon = \sum_{i=1}^n w_i \times \text{is_incorrect}_i$$

$$\alpha = \frac{1}{2} \ln \left(\frac{1 - \epsilon}{\epsilon} \right)$$

Ensemble Learning Methods

Boosting Method: Weight-based Approach

4. **Update Weights:** Increase the weights of the data points the model got wrong. Decrease the weights of the points it got right.

- **For Incorrect Points (The Mistakes):** We multiply their weight by e^{α} . Since α is positive, this **increases** the weight.
 - *Result:* These points become "heavy" and more important for the next model.
- **For Correct Points:** We multiply their weight by $e^{-\alpha}$. This **decreases** the weight.
 - *Result:* These points become "light" because we already know how to solve them.

5. **Repeat:** The next model is forced to pay more attention to the "heavy" (incorrect) points.

Ensemble Learning Methods: Comparison

Method	Strategy	Relation of Models	Primary Goal
Voting	Combine different algorithm types (e.g., KNN + SVM + Trees).	Independent	Improve overall stability.
Bagging	Same algorithm, different random subsets (with replacement).	Independent / Parallel	Reduce Variance (Overfitting).
Pasting	Same algorithm, unique data slices (without replacement).	Independent / Parallel	Reduce Variance (Best for huge data).
Boosting	Same algorithm, models learn from previous errors.	Dependent / Sequential	Reduce Bias (Underfitting).

Thank You

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